

SECR1013 – 03 DIGITAL LOGIC

Project Report Digital Logic

Project Title: Electronic Lift / Elevator Controller System

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Group B: 3-bit Counter (D Flip-flop)



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DEDICATION & ACKNOWLEDGEMENT

This project is dedicated to our lecturer, Dr. Ts. Raja Zahilah binti Raja Mohd. Radzi, for guiding us in the Digital Logic course with patience, knowledge, and encouragement throughout. Without her help and involvement at each step throughout the development of this project, this project and report would have never been done. Her continuous support and guidance have helped us understand the fundamental digital logic concepts, specifically in the designing and implementing of a 3-bit counter using D flip-flops.

We would also like to express our thanks to the Faculty of Computing for providing us the facilities, resources, a learning environment and giving us this great opportunity to work on a real-world-related project.

In addition, we would like to extend our appreciation to all team members for their excellent teamwork, cooperation, and commitment. In order to accomplish this project the best way possible, each member has performed the simulations using Deeds software, prepared the report, and completed the presentation and demonstration video.

Words are not enough to express our heartfelt appreciation and love to our supportive family and friends for their encouragement and support throughout this project. It enabled us to successfully finish this project on time.

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1.0 BACKGROUND

Digital logic systems are the basic framework for modern electronic control systems. Digital logic systems make use of the principles of both combinational logic and sequential logic to develop systems that are able to perform specific tasks. For this project, we will develop a simulation for the Electronic Elevator Controller System based on the basic concepts of digital logic systems, implementing all basic functions, along with security for the particular purpose. Elevators are essential vertical transportation components in taller buildings, requiring reliable control systems to enable user security and ease of use. Although conventional elevator control systems centre on elementary functions such as choosing floors, moving in particular directions, and controlling doors, increasing demand for secure access in home, business, or controlled settings indicates the necessity for unified authentication processes. This particular project utilizes the Deeds Digital Circuit Simulator and addresses a gap between theory and implementation. It reinforces some essential aspects of the Digital Logic course, such as flip-flop circuits, counter circuit designs, comparative logic, and decoder circuit designs. The project showcases how a secure, interactive, and functionally reliable elevator control system can be developed by applying digital logic concepts for performing operations based on required functionalities and integrative security standards.

2.0 PROBLEM STATEMENT

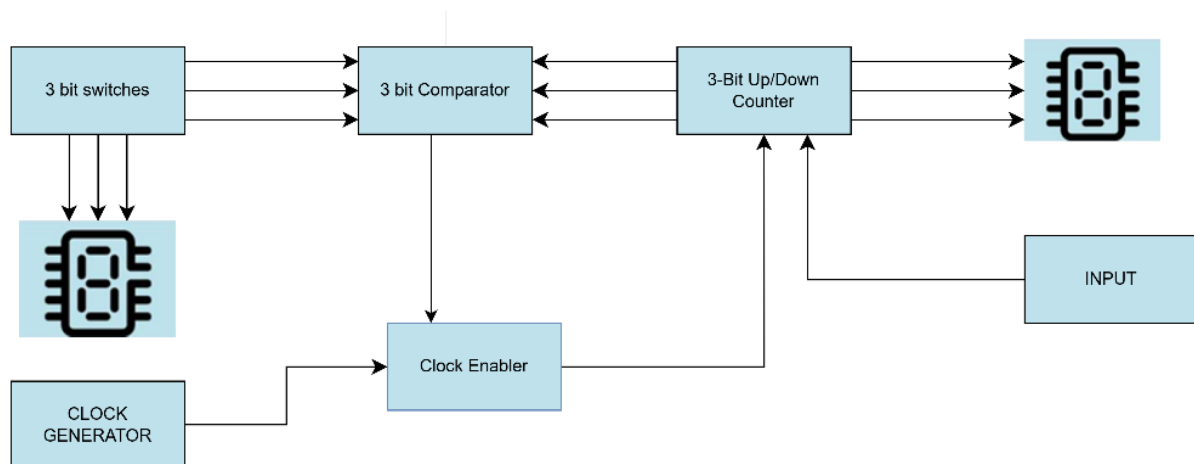
Nowadays, many elevator installations exhibit a significant absence of adequate security mechanisms. This absence poses severe security threats whereby unauthorized people gain access to the use of elevators. The main problem with the current designs is that they do not inspect or check the level of authorization of the user to access particular floors or the elevator itself. As a measure to bridge the security weakness, we propose the development of a new elevator control system using the Digital Electronics Deeds Simulator software. Our proposed system entails the use of passwords for access, with the inclusion of modern digital components to ensure that access to the elevator is limited to authorized individuals only.

3.0 SUGGESTED SOLUTION

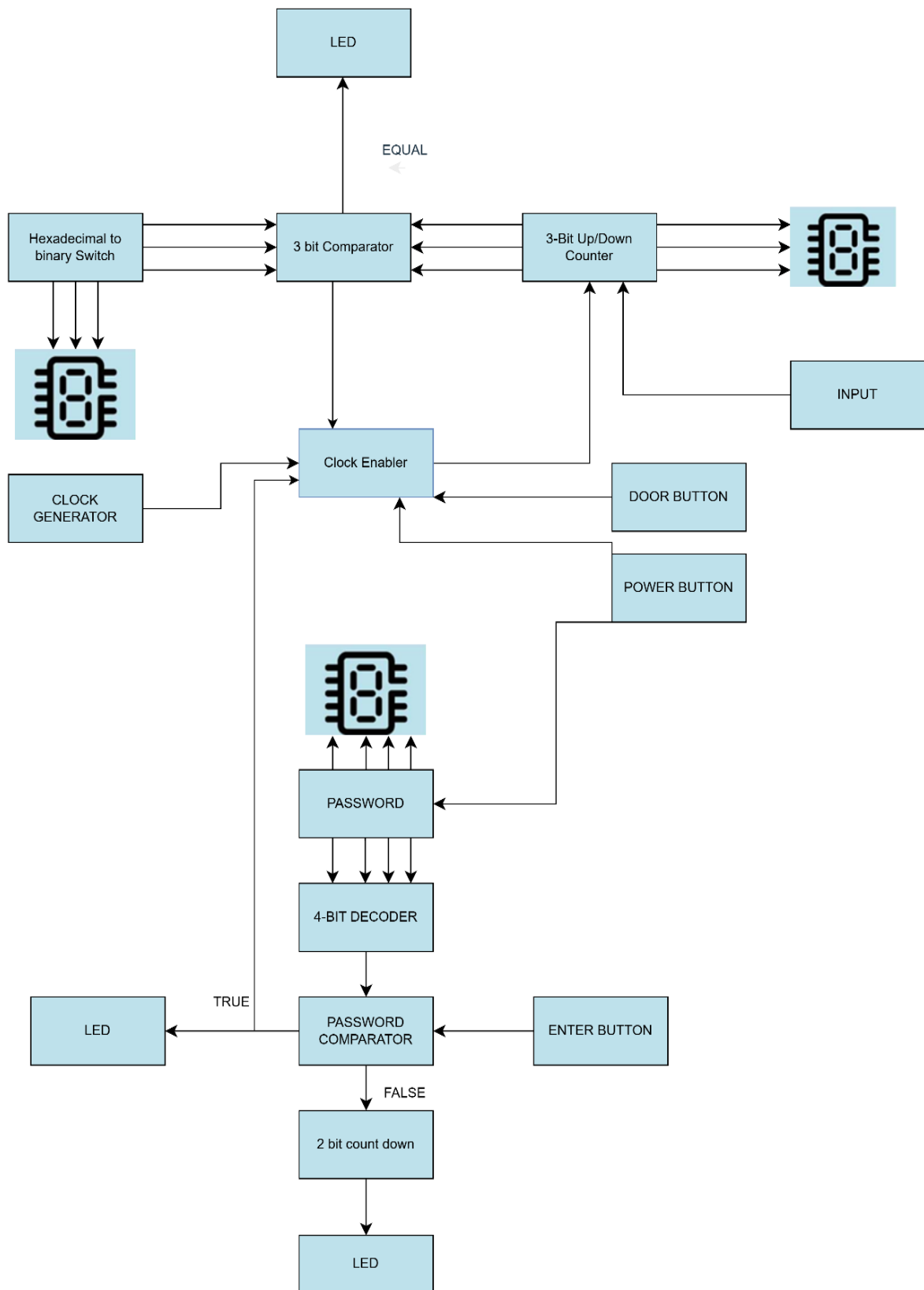
To address the security problem, we propose the implementation of a 4-digit password-protected solution. The proposed solution requires the use of advanced electronic components such as a 3-bit comparator, decoder, basic gates, and D flip-flops. The password hex value will first be entered by the users for them to get access to the lift. The lift will only function after the user has entered the correct password in the password reader. After successful validation and lighting of the LEDs, the user will then change the switch X to 0, meaning going up, and has the option of setting their desired floor level; selecting above 8 levels will make the system display an invalid red LED light. The lift operation will then cease, and LED lights will display when it has arrived at the floor levels selected by the user. The user will then change the switch to 1, meaning going down, and has the option to set their desired floor levels. The proposed solution will give a reliable and efficient password validation process that ensures authorized users are the only ones who can control the lift. The proposed project will be designed by the Digital Electronics Deeds Simulator.

3.1 Block Diagram

Part 1: Basic Lift Controller System

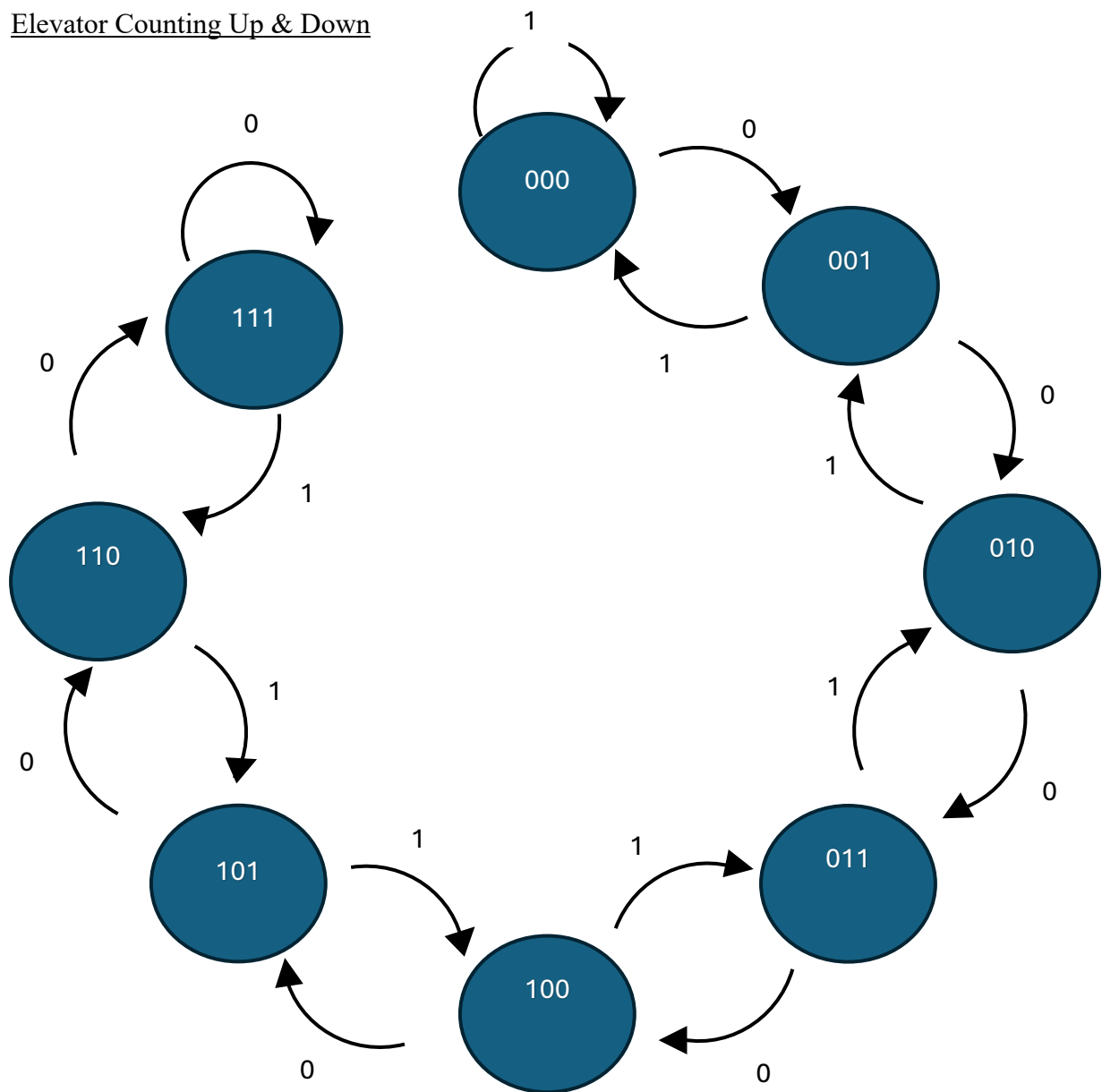


Part 2: Advanced Lift Controller System

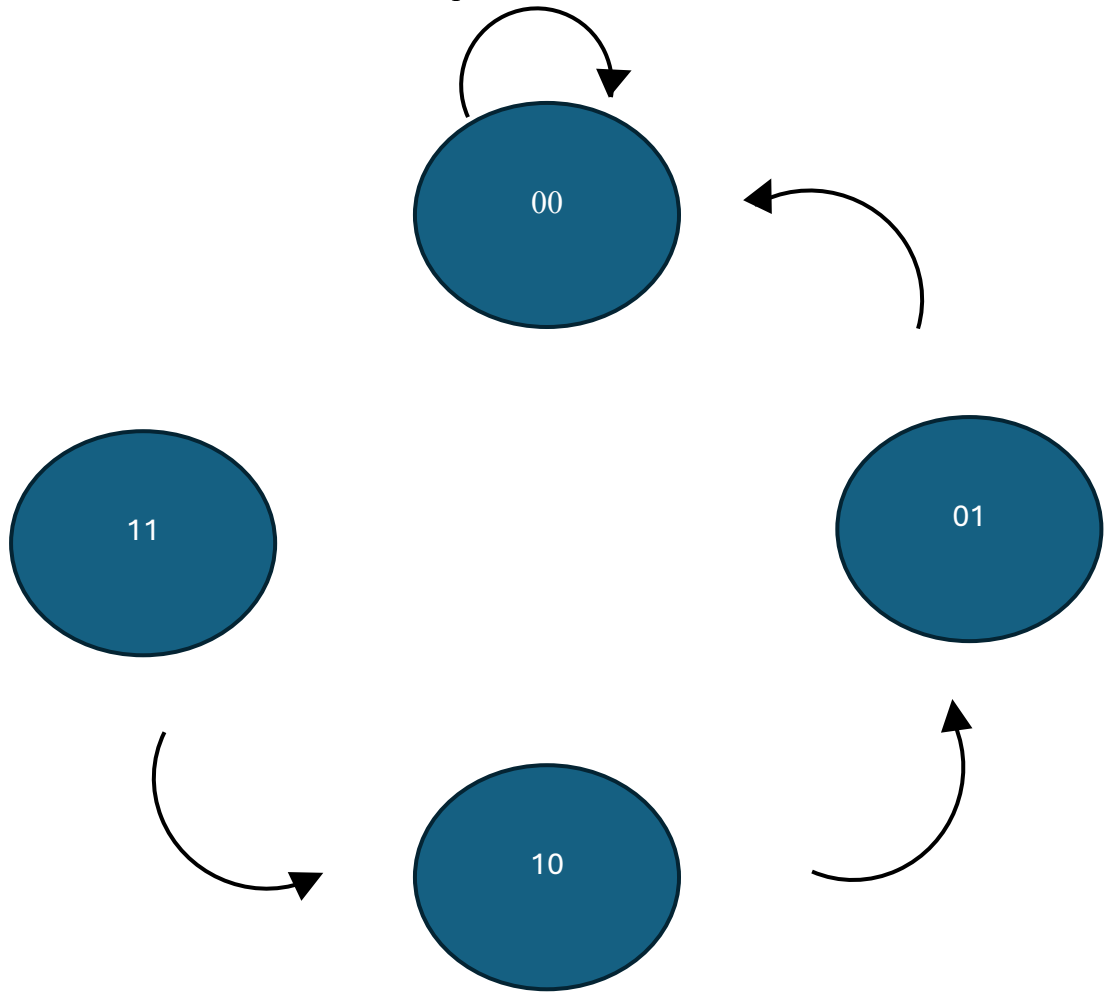


3.2 State Diagram

Elevator Counting Up & Down



Advanced Elevator - 3 Password Attempts



3.3 State Table

Elevator Counting Up & Down

INPUT	PRESENT STATE			NEXT STATE			D FLIP FLOP		
X	Q _C	Q _B	Q _A	Q _{C+}	Q _{B+}	Q _{A+}	D ₂	D _B	D _A
0	0	0	0	0	0	1	0	0	1
0	0	0	1	0	1	0	0	1	0
0	0	1	0	0	1	1	0	1	1
0	0	1	1	1	0	0	1	0	0
0	1	0	0	1	0	1	1	0	1
0	1	0	1	1	1	0	1	1	0
0	1	1	0	1	1	1	1	1	1
0	1	1	1	1	1	1	1	1	1
1	0	0	0	0	0	0	0	0	0
1	0	0	1	0	0	0	0	0	0
1	0	1	0	0	0	1	0	0	1
1	0	1	1	0	1	0	0	1	0
1	1	0	0	0	1	1	0	1	1
1	1	0	1	1	0	0	1	0	0
1	1	1	0	1	0	1	1	0	1
1	1	1	1	1	1	0	1	1	0

Advanced Elevator – 3 Password Attempts

PRESENT STATE		NEXT STATE		D FLIP FLOP	
Q _B	Q _A	Q _{B+}	Q _{A+}	D _B	D _A
0	0	0	0	0	0
0	1	0	0	0	0
1	0	0	1	0	1
1	1	0	0	1	0

3.4 Karnaugh Map (K-Map)

Elevator Counting Up & Down

D₂:

$Q_B Q_C$ $X Q_A$	00	01	11	10
00	0	0	1	0
01	1	1	1	1
11	0	1	1	1
10	0	0	0	0

$$D_2 = \bar{X}Q_C + Q_CQ_A + Q_CQ_B + \bar{X}Q_BQ_A$$

D₁:

$Q_B Q_C$ $X Q_A$	00	01	11	10
00	0	1	0	1
01	0	1	1	1
11	1	0	1	0
10	0	0	1	0

$$D_1 = \bar{X}\bar{Q}_BQ_A + Q_CQ_BQ_A + XQ_BQ_A + \bar{X}Q_B\bar{Q}_A + XQ_CQ_BQ_A$$

D₀:

$Q_B Q_C$ $X Q_A$	00	01	11	10
00	1	0	0	1
01	1	0	1	1
11	1	0	0	1
10	0	0	0	1

$$D_0 = Q_B\bar{Q}_A + \bar{X}\bar{Q}_A + Q_C\bar{Q}_A + \bar{X}Q_CQ_A$$

Advanced Elevator - 3 Password Attempts

D_B:

Q_B \ Q_A	0	1
0	0	0
1	0	1

$$D_B = Q_B Q_A$$

D_A:

Q_B \ Q_A	0	1
0	0	0
1	1	0

$$D_B = Q_B \overline{Q_A}$$

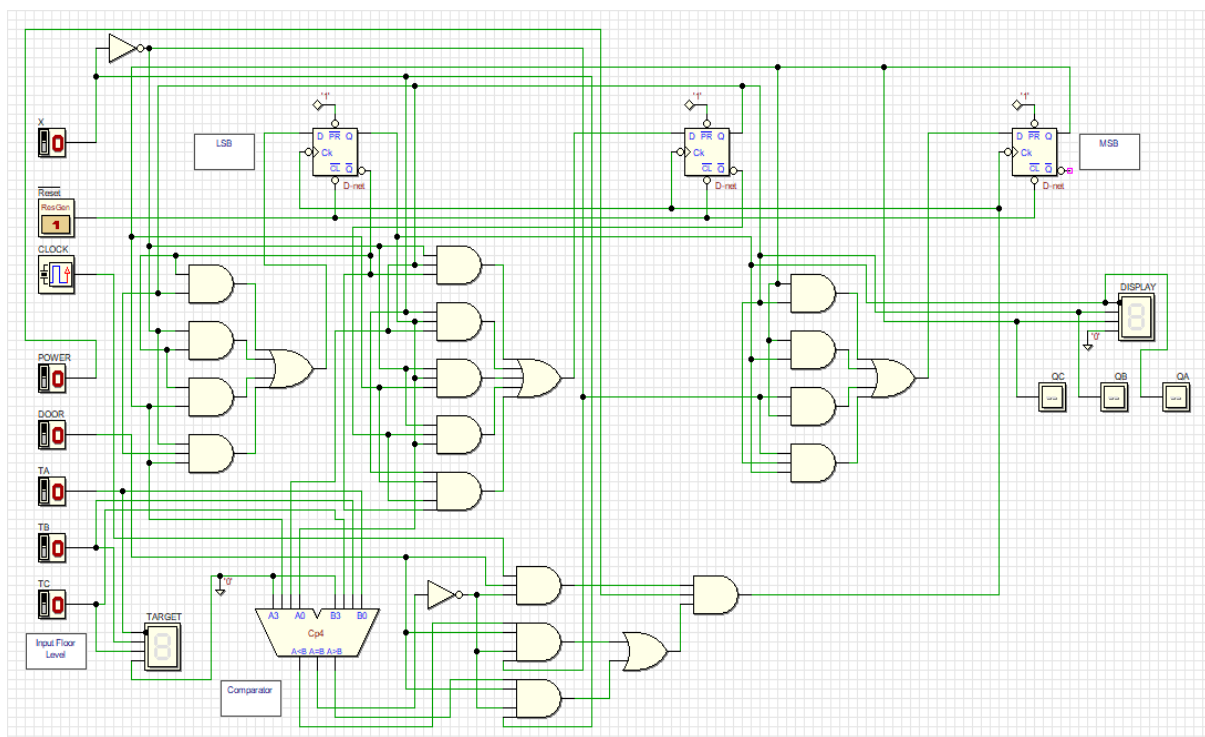
4.0 SYSTEM REQUIREMENTS

The system must fulfil several functional requirements to operate effectively. It needs to support an 8-floor building using 3-bit addressing, provide both up and down movement control with automatic stopping at selected floors, simulate door open and close operations, display current floor numbers clearly, implement 4-digit password authentication with appropriate security measures, and provide visual indicators for system status. Other security features, like attempt counters and system lockout features, may be added as added features in the future. From a technical perspective, the design must make use of D flip-flops for all the sequential logic components, perform all the required simulations using the Deeds software, design a friendly user interface for the simulation and demonstration of the product, and include documentation of the design processes as well as the operations of the product.

5.0 SYSTEM IMPLEMENTATION

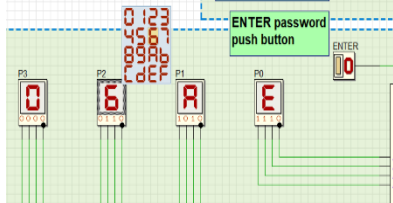
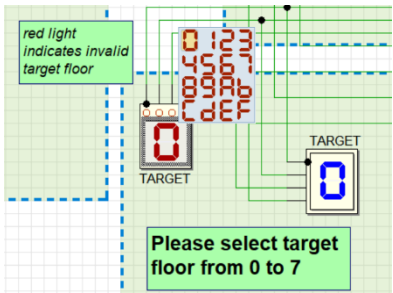
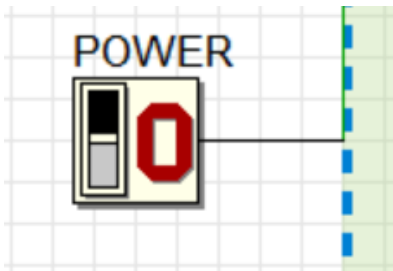
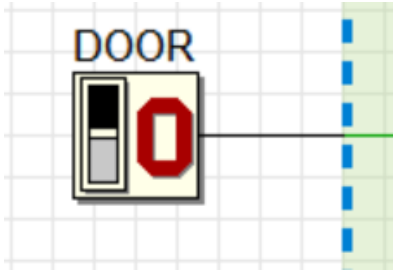
Part 1: Basic Lift Controller System

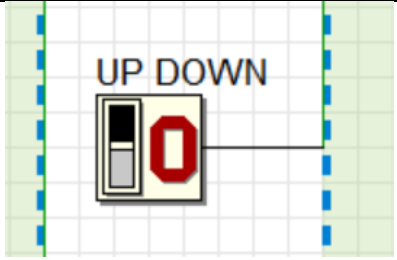
A 3-bit synchronous counter based on negative-edge D flip-flops serves as the core sequencing element, counting from 000 to 111 to represent floors 0 through 7 while providing counting up and down capabilities with reset functionality to another floor. A 3-bit comparator continuously monitors the system state by comparing the current floor position from the counter output with the desired floor input from user selection, utilising comparators for equality checking and generating a low output signal when these values match before connecting to the clock of the negative-edge D flip-flop. The clock enabler functions as a pattern detector that intelligently controls system timing. This component enables the clock signal only when the elevator requires movement and disables it upon reaching the destination. A 7-segment display driver for clear numerical presentation. Initially, the user selects a destination floor using the 3-bit input mechanism, after which the system verifies door closure status. Upon confirmation, the clock enables, the door closes (1), and the counter initiates counting operations while the comparator continuously monitors the current position relative to the destination. When equality occurs between current and destination values, the clock disables automatically, and the door open signal activates while the floor number displays clearly on the 7-segment display. This sequence ensures reliable and predictable elevator operation.



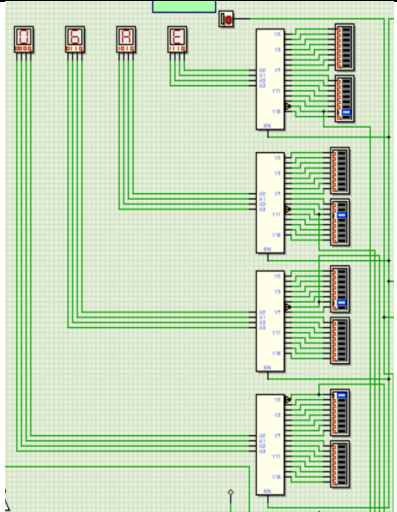
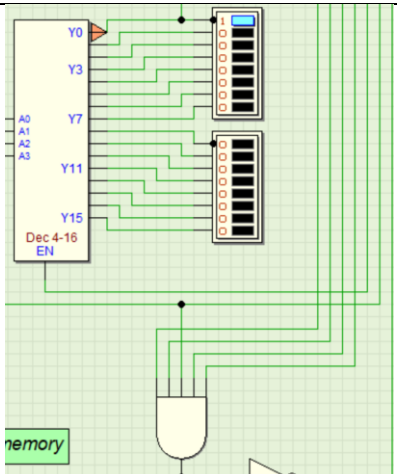
Part 2: Advanced Lift Controller System

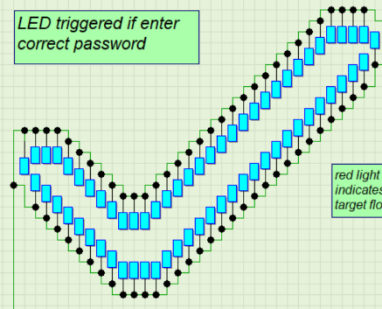
1. Input Switch System

Features Overview	Description	Implementation
Password Input	Hexadecimal input allows users to key in a 4-digit hexadecimal password, with each digit represented by four binary inputs. This configuration enables password values ranging from 0 to F.	
ENTER button	After the user keys in the password, press the ENTER button to confirm.	
Floor Selection	Hexadecimal inputs enable users to select their desired destination floor, with values ranging from 0 to 7. If users select inputs above 7, a red LED will be activated, showing that it is invalid.	
POWER Switch	Power Off (0) Power On (1)	
DOOR Switch	Door Open (0) Door Close (1)	

UP DOWN switch	Going up (0) Going down (1)	
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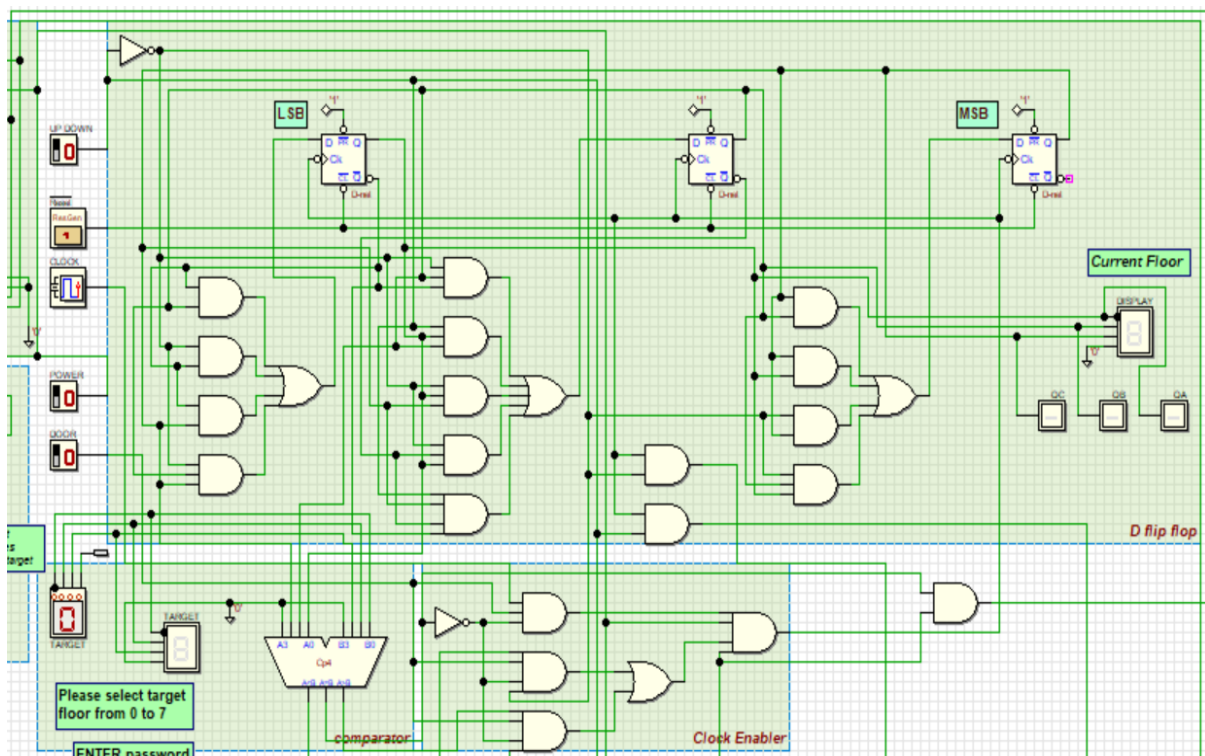
2. Decoder Implementation

Features Overview	Description	Implementation
4-16 decoders	The system employs four 4-16 decoders to process the hexadecimal password input. Each decoder corresponds to one digit of the 4-digit password and converts hexadecimal input into corresponding binary outputs. The enable pin of each decoder is connected to the POWER switch, ensuring the authentication system activates only when the elevator is powered on. Decoder outputs connect to LED arrays that provide visual confirmation of each entered digit.	
	These outputs are fed into an AND gate configured for password comparison.	

Blue tick LED	When all four password digits match the prestored values and the ENTER button is pressed, the final AND gate will activate the blue tick LED, which signals successful authentication and enables the user to proceed to floor selection.	
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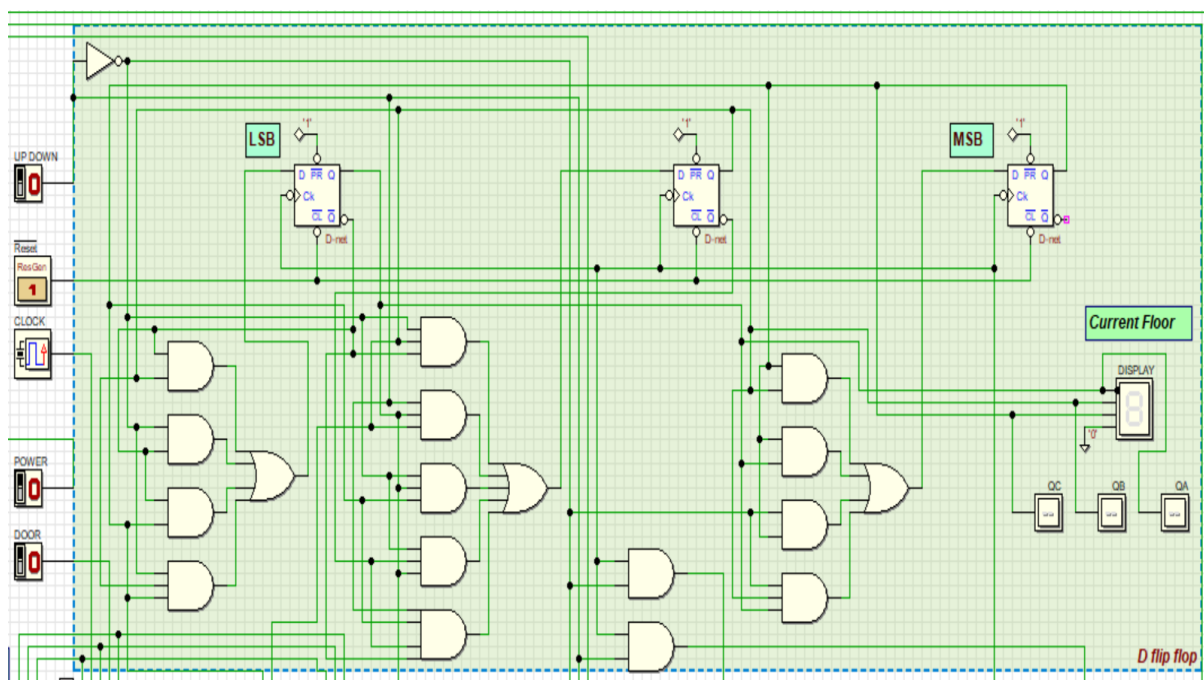
3. Clock Enabler Circuit

The clock enabler circuit is implemented using a 5-input AND gate. The inputs to this gate include the CLOCK signal, the POWER switch signal, the DOOR switch signal, the password authentication signal, and the comparator output signal. The clock enabler becomes active only when all five inputs are in the logic-high (1) state. The output of the 5-input AND gate is connected to the clock input of the D flip-flops. This configuration allows the elevator operation to start or stop based on system conditions. Specifically, the clock signal is disabled when the selected floor has been reached, thereby halting further counting and movement.

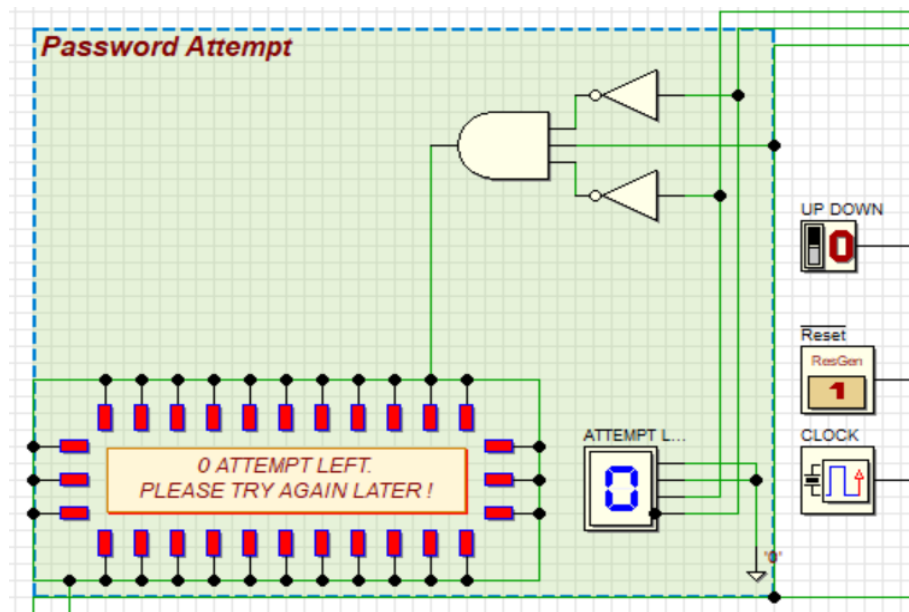


4. D Flip-Flop Sequential Logic

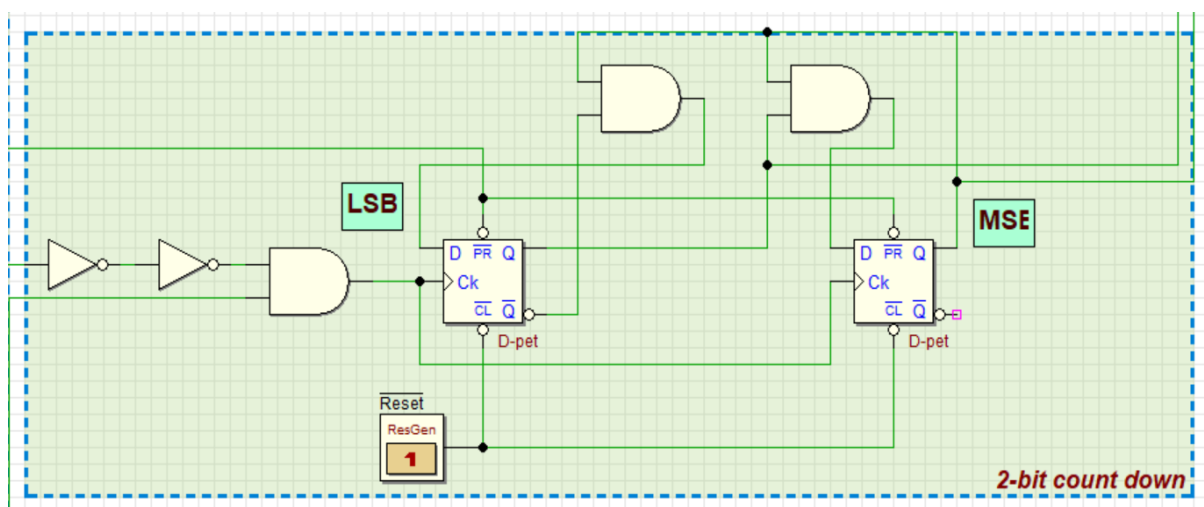
It is implemented using negative-edge triggered D flip-flops as a synchronous 3-bit up and down counter. The mode of operation for the counter is decided by the direction switch logic; 0 moves it upward, while logic 1 moves it down. D inputs are obtained from the next-state equations developed by K-map simplification on the state table. It receives clock pulses only when enabled through the clock enabler circuit. There is also an asynchronous reset to quickly go back to a ground floor (000) if needed. The counter outputs represent the current position regarding the floor and are passed onto the display system and the comparator circuit for real-time tracking of position.



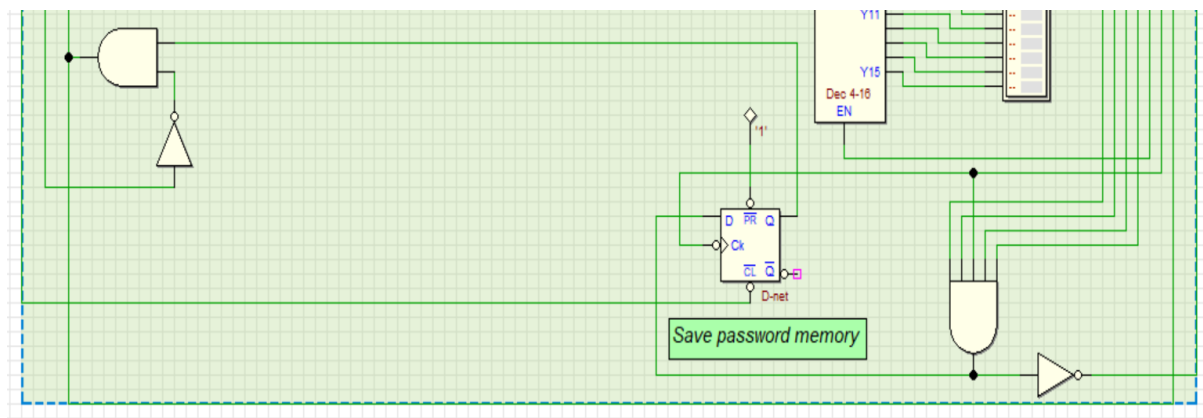
A “3 Password Attempts” module is also implemented in the system, where users will only have 3 attempts to enter the correct password. If they fail to enter the correct password within 3 attempts, a warning alarm will be given, and they will be fined a certain amount of money (in a real-world scenario). Limiting password attempts can prevent brute-force attacks, a security technique where attackers use automated tools to try thousands of password combinations to gain unauthorised access to an account.



The 3 password attempts module works by using two D flip-flops as a synchronous 2-bit down counter, starting from 111 (3) to 000. With the password authentication output and the ENTER button acting as the clock input to the D flip-flops, if the password entered by the user does not match with the prestored values, the clock will be enabled, and the counter counts 1 bit down. The process repeats until the user enters the correct password or the counter reaches 0.

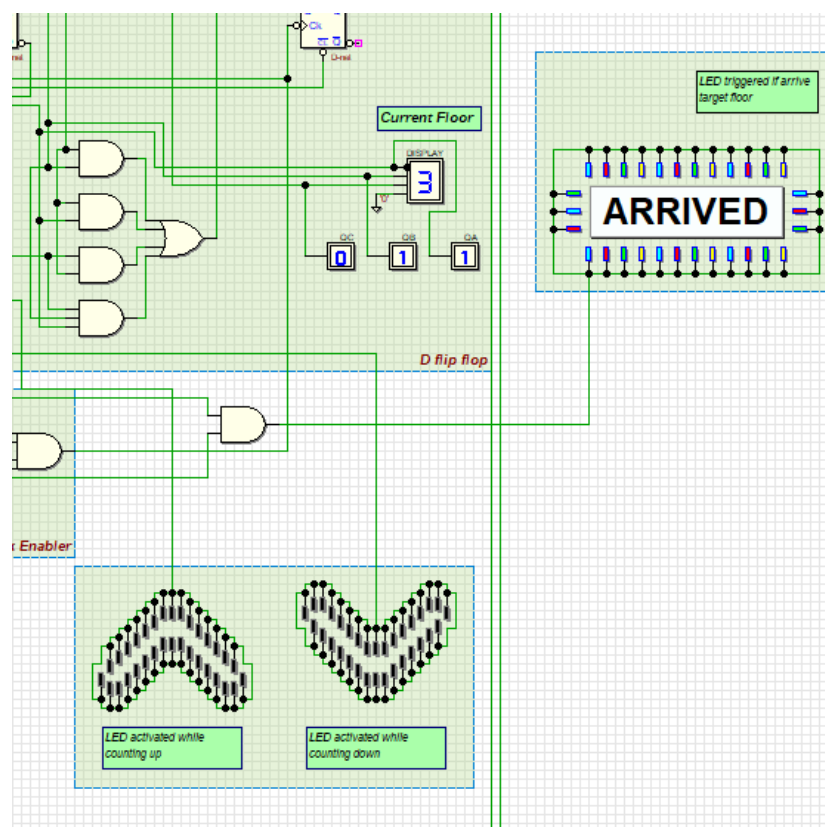


An additional D flip-flop is used to store the memory of the password. Once the ENTER button is pressed, and if all four password digits match the prestored values, the flip-flop sets Q to 1, which then triggers the blue tick LED.



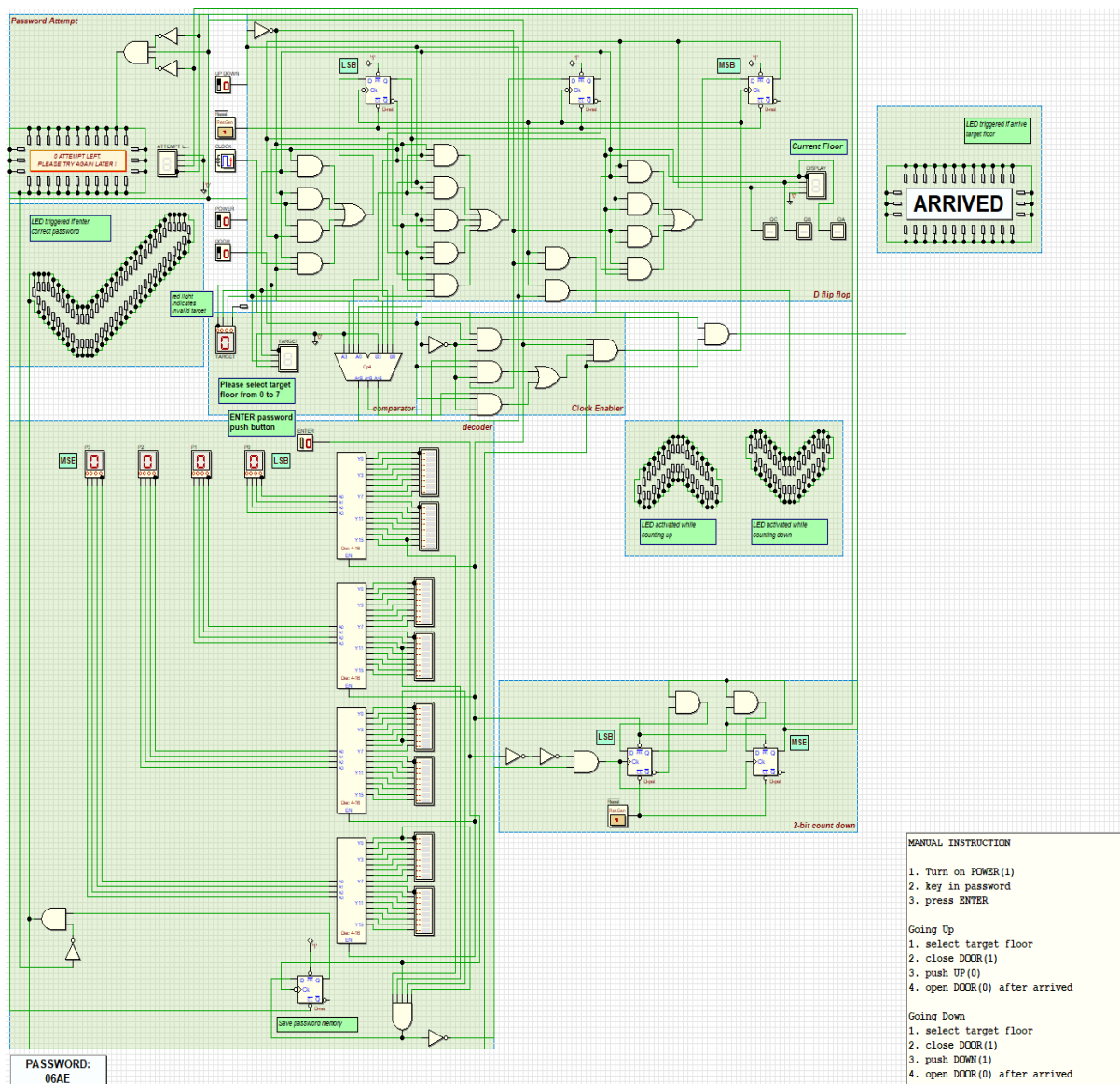
5. Seven-Segment Display System

The displays utilise BCD to 7-segment decoders, which convert binary inputs into proper segment activation patterns. During the ascending of the elevator, the upward arrow LED blinks, whereas if the elevator is descending, the downward arrow LED blinks, providing real-time feedback of the elevator movement. Upon arriving at the destination floor, the ARRIVED alarm will sound to notify the users that they have reached the destination floor.



6. Operational Sequence Integration

To complete the system, a structured operational sequence is followed. After successful password authentication, indicated by the illumination of the blue LED, the user selects the desired floor. Invalid floor selections (values above 7) activate a red warning LED and prevent further system operation. For valid floor selections, the door closure sequence is enabled, followed by direction selection and movement initiation. During travel, the up or down counter increments or decrements toward the destination floor, while the comparator continuously monitors the current position. When the current floor equals the destination floor, the comparator signals the clock disabler to halt the counting process. The system then activates arrival indicators and displays the final floor position and activates the ARRIVED indicator LED to provide a clear visual confirmation of arrival, completing the elevator operational cycle.



6.0 CONCLUSION AND REFLECTION

6.1 Conclusion

A secure electronic elevator controller system was successfully designed and implemented using digital logic in this project. The integration of both basic elevator functionality and advanced security features was achieved through a 4-digit password authentication mechanism. Through the use of D flip flop as 3-bit synchronous counters for controlling each floor and comparator logic for comparing destination, a functional simulation for the elevator controller has been developed. The inclusion of password protection with attempts is reflective of the current concerns for security in the actual operation of the elevators. Sequential circuit logic concepts and decoders, which were examined in the Digital Logic course can also be realized from the design and implementation of a secure electronic system for an elevator controller on the DEEDS software.

6.2 Reflection

Group Member	Reflection
Lim Li Jing	I was able to contribute towards developing a secure and functional elevator controller using digital logic. My achievements included implementing the password authentication module and troubleshooting timing issues during system integration with my teammate. My key strength was translating theoretical knowledge into practical circuits, particularly with decoders and sequential logic. I also contributed effectively to documentation and team coordination. Initially, I struggled with debugging complex state interactions took longer than anticipated and also weak in theoretical knowledge about synchronous and saturated counter but after I done this project I already master in this chapter. To improve the design, I propose adding biometric authentication, energy-saving idle modes, and a priority-based request queue for multi-user scenarios.
Lim Li Ning	I realized that how the elevator controller used in real life through doing the project by using deeds app. Without doing project I didn't understand that either d flip-flop, T flip-flop and J-K flip-flop can use at same time or not. Now, I more deeply notice that both can be used at

	same project, but the connection of the flip-flop must be correct based on your idea. Some of the achievement I did in the project is implementing the first part draft which is counting the up down of the elevator controller and doing comparator and doing the integration with my teammate and checking each of the connection isn't correct or not. However, I was having difficulty with the debugging procedure of complex state interaction. It took longer than expected. To doing this project, I apply theoretical concepts to circuit design regarding decoders and sequential logic. The feature of the project can add on gesture control for elevators, voice interaction, system health monitoring and the predictive maintenance for elevators.
Ng Xuan Yee	This project has taught me how the concepts of digital logic are used in a real-life system, and I feel that many things that I know are like the surface of the sea. Achieving in this project involves understanding the working of D flip-flops and implementing the count sequence correctly. Even though it took me time to debug my circuit, with persistent practice and simulations, my skills in this area improved. This project has increased my confidence level in the design and simulation of sequential circuits. In my opinion, the design of the controller either in real-world components or in an FPGA setup such as in this project would have provided more practical knowledge about the robustness of the design not only reliant on software simulation.
Cheng Zhi Min	Throughout this project, my understanding about the application of concepts in digital logic became clearer, especially about flip-flops and counters. Some of the achievements include the implementation of the password limit feature to inhibit malicious activities. Moreover, the final design can still achieve improvement with the implementation of a programable password feature. The feature will enable the user to change the password anytime they want rather than having a fixed password. Overall, implementing this project has increased my knowledge in digital electronics largely despite the challenges involved in the connections, as well as debugging processes.

7.0 REFERENCES

Digital Logic (5th ed.). (2024). Faculty of Computing, Universiti Teknologi Malaysia.

Mohd Fo'ad bin Rohani. (2025). Project Digital Logic: Electronic Lift/Elevator Controller System (Simulation Using Deeds Software). Faculty of Computing, Universiti Teknologi Malaysia.

8.0 APPENDICES

8.1 Digital Circuit Simulator (DEEDS)

Part 1: [Click here](#)

Part 2: [Click here](#)

8.2 DEEDS Simulation Demonstration Video

[Click here](#)

8.3 Presentation Slide

[Click here](#)

8.4 Presentation Video

[Click here](#)

8.5 Group Discussion Evidence



8.6 Task Distribution

No.	Name	Matric No.	Task Description
1.	Lim Li Jing	A25CS0248	• DEEDS Circuit Implementation • Report Writing • Slide Preparation • Video Presenting
2.	Lim Li Ning	A25CS0249	• DEEDS Circuit Implementation • Report Writing • Slide Preparation • Video Presenting
3.	Ng Xuan Yee	A25CS0291	• DEEDS Circuit Implementation • Report Writing • Slide Preparation • Video Presenting
4.	Cheng Zhi Min	A25CS0050	• DEEDS Circuit Implementation • Report Writing • Slide Preparation • Video Presenting